

Product Requirements Document (PRD): Holistic Learning Ecosystem

****For QA & Testing****

1. Introduction

****Purpose:**** This document outlines the existing features, workflows, and platform capabilities of the Holistic Learning Ecosystem. It is designed to serve as the single source of truth for QA engineers and testers to understand what has been built, the intended behavior of the system, and what needs to be validated across the web and mobile platforms.

2. Platform Overview

The Holistic Learning Ecosystem is a comprehensive EdTech platform with three main access points:

1. ****Frontend Web Application (Next.js):**** For Students and Instructors.
2. ****Mobile Application (React Native):**** For Students on-the-go.
3. ****Backend API (FastAPI):**** Centralized REST API for all data and logic.

User Roles

- ****Student:**** Can browse, enroll, learn, earn coins/achievements, participate in discussions, and attend live classes. Access is gated by customized permissions (e.g., specific access to Batch 1, Batch 2, RAS, Graphotherapy).
- ****Instructor:**** Can create/manage courses, upload content, monitor student progress, and view revenue analytics. Can manage specific student cohorts (e.g., Batch 1 Manager).
- ****Administrator (Implicit):**** Manages global settings, platform analytics, email templates.

3. Core Features & Testable Scenarios

3.1. Authentication & Authorization (Web & Mobile)

- ****What is built:**** JWT-based secure login with Role-Based Access Control (RBAC). A complex feature-flag mapping (`USER\_ACCESS\_MAP`) restricts route access per individual user.
- ****To be tested:****
 - [] User Registration (Student & Instructor roles).
 - [] Login/Logout flows on Web and Mobile.
 - [] Password reset and session expiration.
 - [] ****RBAC & Feature Flags:**** Ensure users only see dashboards they are authorized for (e.g., RAS students cannot see Batch 1 polity, Upanishad students cannot see RAS PYQs).

3.2. Specialized Domain Learning Modules (CRITICAL)

- ****What is built:**** Tailored educational environments for distinct groups (UPSC, RAS, Ancient Wisdom, Graphotherapy, Meditation).
- ****To be tested:****
 - [] ****Batch 1 & 1.1 (UPSC Preparation):****
 - Verify access to Daily/Weekly schedules (`batch1-1` portal).
 - Test the ****Evening Session**** functionality, specifically the customized ****Pomodoro Timers**** tied to Saturday Tests and Daily Cycles.

- Validate subject-specific modules: Polity (Flashcards, MCQs, High Court module, Coalition Govt), History (Ancient Types), Environment, Geography, and Science & Tech.
- Test the centralized **Question Bank** and **Current Affairs** dashboard access.
- Check the customized Revision Portal, QA Test History Modal, and Batch 1-specific Leaderboard/Analytics.
- Test the **UPSC Store** access for purchasing and unlocking specific content plans.
- Test the embedded **AI Tutor / AI Coach** functionalities across the revision subjects.
- [] **Batch 2 (Ancient Wisdom / Sadhana):**
 - Check the **Sadhana Portal V5** flows (Sadhana Journey View, Session Timer).
 - Validate Ancient Wisdom content delivery: Upanishads (Kena, Amritanada, Aruni, etc.) and Knowledge Graph visualizations.
 - Test out the dedicated **Productivity** tools associated with daily Sadhana tracking.
 - Test the specific Subscription Gate that protects Batch 2 modules.
- [] **RAS (Rajasthan Administrative Service) Prep:**
 - Verify the RAS specific portal (`rasRevision``).
 - Test RAS specific viewers: Syllabus Viewer, PYQ Viewer (Previous Year Questions), and Science & Tech Viewer.
 - Validate RAS MCQ Test Session accuracy and tracking, along with the distinct RAS Pomodoro Timer.
- [] **Ancient Graphotherapy:**
 - Verify the standalone Graphotherapy Dashboard.
 - Test the Graphotherapy checkout/onboarding funnel.
- [] **Meditation Portal:**
 - Test the dedicated Meditation dashboard, validating multimedia playback, session tracking, and user progress recording without interruptions.

3.3. Extended Management Portals

- **What is built:** Distinct, isolated dashboards for staff members operating outside the standard Instructor scope.
- **To be tested:**
 - [] **Admin Portal**
`((dashboard)/admin)(file:///d:/Graphology/Master%20Software/Eduecosystem/frontend/src/config/user-access-config.ts#260-267):`
 - Global system configurations, UPSC Plan/Bundle management, Development Action Logs, PDR tracking, and Daily Reports overview.
 - [] **CRM Portal**
`((dashboard)/crm)(file:///d:/Graphology/Master%20Software/Eduecosystem/frontend/src/config/user-access-config.ts#260-267) & [(mobile-crm)](file:///d:/Graphology/Master%20Software/Eduecosystem/frontend/src/config/user-access-config.ts#260-267):`
 - Lead management, student profiling, pipeline funnels, and mobile-optimized CRM views for sales staff.
 - [] **Marketing Tools**
`((marketing))(file:///d:/Graphology/Master%20Software/Eduecosystem/frontend/src/config/user-access-config.ts#260-267) & [(dashboard)/marketing](file:///d:/Graphology/Master%20Software/Eduecosystem/frontend/src/config/user-access-config.ts#260-267):`

- Announcement broadcasting, marketing wallet/transactions tracking, public landing page controls, and user acquisition funnels.

3.4. Instructor / Teaching Workflow (Web Only)

- **What is built:** A complete suite for course creators and cohort managers.
- **To be tested:**
 - [] **Course Creation & Management:** 4-step wizard with strict form validation, drag & drop module organization (`@dnd-kit`).
 - [] **Cohort Management:** Validate the specific `batch1-manager` dashboard for monitoring Batch 1 student progress (`MorningBriefing`, `AtRiskDetector`, `CohortComparison`).
 - [] **Analytics & Revenue:** Real-time earnings tracking, enrollment counts, and student progress metrics via charts.
 - [] **Announcements & Communication:** Broadcast notifications (`CommunicationHub`) to specific cohorts.

3.5. Student Learning Experience (General)

- **What is built:** The primary unified learning interface.
- **To be tested:**
 - [] **Course Discovery:** Advanced search with multi-filters.
 - [] **Video Player:** Full-featured playback controls, tracking watch time.
 - [] **Assessments:** Quizzes and peer-reviewed assignments.
 - [] **Study Utilities:** Creating rich text notes, bookmarking.
 - [] **Certificates:** Auto-generation upon 100% completion.

3.6. Gamification System (Web & Mobile)

- **What is built:** A robust engagement loop spanning the entire platform.
- **To be tested:**
 - [] **Coin Triggers:** Verify coins are awarded automatically for 17 actions (lesson completion, Pomodoro sync, daily streak, discussion posts, etc.).
 - [] **Achievements:** Unlock badges correctly across 6 categories.
 - [] **Leaderboard:** Ranks update accurately based on earned XP/Coins (ensure isolation if leaderboard is cohort-specific).

3.7. Communication & Community (Web & Mobile)

- **What is built:** Interactive socialization elements.
- **To be tested:**
 - [] **Discussion Forums:** Threads, nested replies, upvoting.
 - [] **Live Classes (Web / Mobile via Agora):** Chat, emoji reactions, live polls, Q&A.
 - [] **Peer Reviews:** Student submission of structured feedback.

3.8. Notification Systems

- **What is built:** Extensible email and in-app alerts.
- **To be tested:**
 - [] **Toast Notifications:** UI alerts appearing for generic actions.
 - [] **Email Preferences & Delivery:** Ensure specific notifications fire for course enrollment confirmations, assignment grading, and quiz results.

4. Non-Functional Requirements (NFRs) to Validate

4.1. UI/UX & Responsive Design

- **Requirement:** WCAG component accessibility, screen reader support, mobile-first design.
- **Test:**
 - [] Test UI layout degradation on Desktop, Tablet, and Mobile browser breakpoints (Next.js web). Check all customized dashboards (RAS, Batch 1, Graphotherapy).

4.2. Performance & Error Handling

- **Requirement:** Fast load times (<2s initial), API constraints (100req/min).
- **Test:**
 - [] Force front-end errors to assert graceful error boundary loading. Verify UI skeleton loaders appear during data fetching.

5. Technical Context for Test Automation

If QA is writing automation tests (e.g., Cypress, Playwright, Detox):

- **Frontend App Structure:** Built with Next.js 16 (App Router). Crucial routing paths to script against include ``/(student-portal)/student/batch1``, ``/(student-portal)/student/batch2``, ``/ras``, ``/upsc``, ``/graphotherapy``, and ``/meditation``.
- **State Management:** Uses Jotai and Zustand. Tests involving the Pomodoro timers must wait or mock time to validate session completion logic.
- **Feature Flags:** Automation scripts must use specific test users mapped in ``USER_ACCESS_MAP`` (e.g. testing ``chitrakumawat33@gmail.com`` for RAS access, ``kajaldhannatar@gmail.com`` for Batch 1).